

The SYK Antipode Theorem:

An Exact Operator Identity for the 14 Torus Modes at $\lambda = 4$

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

October 2026

(Revised: response to peer review)

Abstract

We prove the SYK Antipode Theorem: for the 8×8 torus at $\lambda = 4$, the bilinear operators constructed from antipodal torus modes satisfy the exact operator identity

$$\mathcal{O}_{A(m)} = -\mathcal{O}_m$$

for every realization of the SYK₄ ensemble. Consequently, the spectral weights satisfy $w_m = w_{A(m)}$ exactly for every state, every realization, and every order in $1/N$. The proof is by direct evaluation of the bilinear coefficients: because the antipode map acts as complex conjugation on the plane-wave basis, the coefficients satisfy $c_{ab}^{A(m)} = -c_{ab}^m$ identically, and the operator identity follows from antisymmetry of Majorana bilinears. No ensemble average is required, and no phase ambiguity arises: the sign is exactly -1 . We also prove the geometric uniqueness of the antipode map and correct an earlier claim of a per realization visible/hidden labeling. The result is verified numerically across 10 disorder realizations, where the equality holds to all displayed digits. We scope the “no $1/N$ correction” claim precisely: it applies to the specific bilinear two-point functions under consideration, and is a consequence of the exact operator identity, not a general non-renormalization statement about other correlators.

Contents

1	Introduction	2
1.1	What is proven	2
1.2	What is corrected	2
1.3	Organization	3
2	Geometry of the Antipode Map	3
2.1	The torus and its Laplacian	3
2.2	The 14-mode set	3
2.3	The antipode map	3
2.4	Uniqueness of A	4
2.5	Geometric characterization	4
2.6	Orbit decomposition	4
3	The Exact Operator Identity	4
3.1	Bilinear operators from torus modes	4
3.2	Coefficient sign under antipode	5
3.3	The operator identity	5
3.4	Vanishing bilinears for self-paired modes	6

4 Numerical Verification	6
4.1 Setup	6
4.2 Result	6
4.3 On the conformal dimension of the bilinears	7
5 Discussion	7
5.1 Scope of the “no $1/N$ correction” claim	7
5.2 Why the earlier paper’s claim failed	7
5.3 The physical content of the theorem	7
5.4 Falsification	8
6 Conclusion	8

1 Introduction

The SYK model [1, 2] provides a solvable example of a strongly correlated quantum system with maximal chaos and a holographic dual. When the model is restricted to a discrete substrate the 8×8 torus at $\lambda = 4$ the 14 degenerate modes form a natural bilinear sector whose symmetries reflect the geometry of the torus.

An earlier paper [5] claimed that the visible/hidden labeling of the 14 modes under SYK projection was exactly the antipode map $A(n_1, n_2) = (-n_1 \bmod 8, -n_2 \bmod 8)$, and conjectured a sum rule with an undetermined normalization coefficient c_1 . The present paper corrects this claim and replaces it with a stronger result.

1.1 What is proven

1. **Geometric uniqueness.** The antipode map A is the unique point reflection of $\mathbb{Z}_8 \times \mathbb{Z}_8$ preserving $\mathcal{K}_{14}$ with exactly two fixed points in $\mathcal{K}_{14}$. It acts as complex conjugation on the plane wave basis of the torus Laplacian. (Section 2.)
2. **Exact operator identity.** For every mode $m \in \mathcal{K}_{14}$ and every embedding ι of Majorana indices into torus sites, the bilinear operators satisfy $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ as exact operator identities, independent of the SYK Hamiltonian and independent of the realization. (Theorem 3.5.)
3. **Spectral weight equality.** Consequently, for every realization of the SYK₄ Hamiltonian and every state $|\psi\rangle$, we have $\langle\psi|\mathcal{O}_m^\dagger\mathcal{O}_m|\psi\rangle = \langle\psi|\mathcal{O}_{A(m)}^\dagger\mathcal{O}_{A(m)}|\psi\rangle$ exactly. (Corollary 3.6.)
4. **No $1/N$ correction to the sum rule.** The equality $w_m = w_{A(m)}$ is exact at all orders in $1/N$. No coefficient c_1 and no marginal kernel derivative k'_c appear. (Corollary 3.7.)

1.2 What is corrected

An earlier paper claimed a specific per-realization visible/hidden labeling based on numerical evidence at $N = 16$, seed 42, row embedding. The present paper shows that *no such labeling can exist*: the operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ forces the two modes of every antipodal pair to have identical expectation values in every state, so neither can be singled out as “visible” over the other.

1.3 Organization

Section 2 establishes the pure-geometry uniqueness of A . Section 3 states and proves the operator identity. Section 4 presents the numerical verification. Section 5 discusses the scope of the result. Section 6 concludes.

2 Geometry of the Antipode Map

2.1 The torus and its Laplacian

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete Laplacian on complex functions $f : \Lambda \rightarrow \mathbb{C}$ is

$$(\Delta f)(n_1, n_2) = 4f(n_1, n_2) - f(n_1 + 1, n_2) - f(n_1 - 1, n_2) - f(n_1, n_2 + 1) - f(n_1, n_2 - 1), \quad (1)$$

with periodic boundary conditions. On the plane-wave basis $v_k(n) = e^{i\pi k \cdot n/4}$ with $k \in \Lambda$, the Laplacian is diagonal:

$$\lambda(k_1, k_2) = 4 - 2 \cos\left(\frac{\pi k_1}{4}\right) - 2 \cos\left(\frac{\pi k_2}{4}\right). \quad (2)$$

2.2 The 14-mode set

Theorem 2.1 (Structure of $\mathcal{K}_{14}$). *The set $\mathcal{K}_{14} := \{k \in \Lambda : \lambda(k) = 4\}$ has exactly 14 elements, given explicitly by*

$$\begin{aligned} \mathcal{K}_{14} = \{ & (0, 4), (4, 0), (1, 3), (1, 5), (2, 2), (2, 6), (3, 1), \\ & (3, 7), (5, 1), (5, 7), (6, 2), (6, 6), (7, 3), (7, 5) \}. \end{aligned} \quad (3)$$

Proof. The condition $\lambda(k) = 4$ becomes $\cos(\pi k_1/4) + \cos(\pi k_2/4) = 0$. Using the sum-to-product identity,

$$2 \cos\left(\frac{\pi(k_1 + k_2)}{8}\right) \cos\left(\frac{\pi(k_1 - k_2)}{8}\right) = 0. \quad (4)$$

The first factor vanishes iff $k_1 + k_2 \equiv 4 \pmod{8}$; the second iff $k_1 - k_2 \equiv 4 \pmod{8}$. Each set has 8 elements; their intersection is $\{(0, 4), (4, 0)\}$, of size 2. Total: $8 + 8 - 2 = 14$. $\square$

2.3 The antipode map

Definition 2.2 (Antipode). The antipode map $A : \Lambda \rightarrow \Lambda$ is

$$A(n_1, n_2) = (-n_1 \bmod 8, -n_2 \bmod 8). \quad (5)$$

Proposition 2.3 (Basic properties). *The map A satisfies:*

- (i) $A^2 = \text{id}$ (involution);
- (ii) A has exactly 4 fixed points on Λ : $(0, 0), (0, 4), (4, 0), (4, 4)$;
- (iii) A has exactly 2 fixed points in $\mathcal{K}_{14}$: $(0, 4)$ and $(4, 0)$;
- (iv) $\mathcal{K}_{14}$ is A -invariant.

Proof. (i) $-(-n) = n$. (ii) $n = A(n)$ iff $2n \equiv 0 \pmod{8}$, i.e. $n_1, n_2 \in \{0, 4\}$. (iii) Of the four fixed points, $(0, 0)$ and $(4, 4)$ have $\lambda = 0$ and 8, so are not in $\mathcal{K}_{14}$. (iv) $\cos(-\theta) = \cos \theta$ gives $\lambda(A(k)) = \lambda(k)$. $\square$

2.4 Uniqueness of A

Definition 2.4 (Point reflection). A point reflection of Λ is a map $T_{a,b}(n_1, n_2) = ((a - n_1) \bmod 8, (b - n_2) \bmod 8)$ for $(a, b) \in \Lambda$.

Theorem 2.5 (Uniqueness of A among point reflections). *Among the 64 point reflections of Λ , exactly two preserve $\mathcal{K}_{14}$: $T_{0,0} = A$ and $T_{4,4}$. Of these, only $T_{0,0}$ has exactly two fixed points in $\mathcal{K}_{14}$.*

Proof. A point reflection $T_{a,b}$ has fixed points iff $2n \equiv (a, b) \pmod{8}$. If a or b is odd, there are no fixed points. If both are even, there are four fixed points. Enumerating the 16 even-even cases and checking closure of $\mathcal{K}_{14}$ under each, only $(a, b) = (0, 0)$ and $(a, b) = (4, 4)$ preserve the set. For $T_{0,0}$, the two fixed points inside $\mathcal{K}_{14}$ are $(0, 4)$ and $(4, 0)$. For $T_{4,4}$, the four fixed points are $(2, 2), (2, 6), (6, 2), (6, 6)$, all inside $\mathcal{K}_{14}$. So only $T_{0,0}$ has exactly two fixed points in $\mathcal{K}_{14}$. $\square$

2.5 Geometric characterization

Theorem 2.6 (A is complex conjugation). *Under the plane-wave basis $v_k(n) = e^{i\pi k \cdot n/4}$, the antipode map satisfies, for every $k \in \Lambda$ and every $n \in \Lambda$,*

$$v_k(A(n)) = \overline{v_k(n)} = v_{A(k)}(n). \quad (6)$$

That is, A is the coordinate expression of complex conjugation on the plane-wave basis.

Proof. $v_k(A(n)) = e^{i\pi(k_1(-n_1) + k_2(-n_2))/4} = e^{-i\pi k \cdot n/4} = \overline{v_k(n)}$. Separately, $v_{A(k)}(n) = e^{i\pi(-k_1 n_1 - k_2 n_2)/4} = e^{-i\pi k \cdot n/4}$. The three expressions agree. $\square$

Corollary 2.7. *A is the unique involution on Λ whose action on the plane-wave basis is complex conjugation.*

Proof. Complex conjugation on the basis $\{v_k\}$ is anti-linear and involutive. Any other anti-linear involution that acts as complex conjugation on all basis vectors would have to agree with A on every v_k , and hence on all of $L^2(\Lambda)$. $\square$

2.6 Orbit decomposition

Theorem 2.8 (Antipode orbits). *The set $\mathcal{K}_{14}$ partitions into 8 orbits under A : 2 fixed points and 6 pairs of size 2.*

Proof. $14 = 2 + 2 \cdot 6$. $\square$

The 8 orbits are:

$$\begin{aligned} &\{(0, 4)\}, \quad \{(4, 0)\}, \\ &\{(1, 3), (7, 5)\}, \quad \{(1, 5), (7, 3)\}, \\ &\{(2, 2), (6, 6)\}, \quad \{(2, 6), (6, 2)\}, \\ &\{(3, 1), (5, 7)\}, \quad \{(3, 7), (5, 1)\}. \end{aligned}$$

3 The Exact Operator Identity

3.1 Bilinear operators from torus modes

Definition 3.1 (Bilinear operator). Fix an embedding $\iota : \{1, \dots, N\} \rightarrow \Lambda$ of Majorana indices into torus sites. For $m \in \mathcal{K}_{14}$, define the bilinear operator

$$\mathcal{O}_m = \sum_{1 \leq a < b \leq N} c_{ab}^m \psi_a \psi_b, \quad c_{ab}^m := 2 \operatorname{Im} \left(v_m(\iota(a)) \overline{v_m(\iota(b))} \right), \quad (7)$$

where $v_m(x) = e^{i\pi m \cdot x/4}$ is the plane wave at mode m .

Remark 3.2 (Antisymmetry). The coefficients c_{ab}^m are real. The operators $\psi_a \psi_b$ for $a \neq b$ satisfy $(\psi_a \psi_b)^\dagger = \psi_b \psi_a = -\psi_a \psi_b$. Therefore $\mathcal{O}_m^\dagger = -\mathcal{O}_m$, i.e., $\mathcal{O}_m$ is anti-Hermitian.

3.2 Coefficient sign under antipode

Lemma 3.3 (Exact sign flip). *For every $m \in \mathcal{K}_{14}$ and every pair $a < b$,*

$$c_{ab}^{A(m)} = -c_{ab}^m. \quad (8)$$

Proof. By definition,

$$c_{ab}^{A(m)} = 2 \operatorname{Im} \left(v_{A(m)}(\iota(a)) \overline{v_{A(m)}(\iota(b))} \right). \quad (9)$$

By Theorem 2.6, $v_{A(m)}(\iota(a)) = \overline{v_m(\iota(a))}$ and $v_{A(m)}(\iota(b)) = \overline{v_m(\iota(b))}$, so

$$c_{ab}^{A(m)} = 2 \operatorname{Im} \left(\overline{v_m(\iota(a))} v_m(\iota(b)) \right). \quad (10)$$

Let $z := v_m(\iota(a)) \overline{v_m(\iota(b))}$. Then $\overline{v_m(\iota(a))} v_m(\iota(b)) = \bar{z}$, and $\operatorname{Im}(\bar{z}) = -\operatorname{Im}(z)$. Therefore

$$c_{ab}^{A(m)} = -2 \operatorname{Im}(z) = -c_{ab}^m. \quad \square \quad (11)$$

□

Remark 3.4 (No phase ambiguity). The sign in Lemma 3.3 is exactly -1 , not a phase $e^{i\phi}$ for some ϕ . This follows from the fact that the coefficients c_{ab}^m are real by construction (they are twice the imaginary part of a complex number), and complex conjugation of a real quantity is itself.

3.3 The operator identity

Theorem 3.5 (SYK-Antipode Theorem). *For every mode $m \in \mathcal{K}_{14}$ and every embedding ι ,*

$$\boxed{\mathcal{O}_{A(m)} = -\mathcal{O}_m} \quad (12)$$

as an exact operator identity, independent of the SYK Hamiltonian and independent of the realization of the random couplings.

Proof. Using Definition (7) and Lemma 3.3,

$$\mathcal{O}_{A(m)} = \sum_{a < b} c_{ab}^{A(m)} \psi_a \psi_b = \sum_{a < b} (-c_{ab}^m) \psi_a \psi_b = - \sum_{a < b} c_{ab}^m \psi_a \psi_b = -\mathcal{O}_m. \quad (13)$$

The identity is exact and does not depend on H or on J_{ijkl} . □

Corollary 3.6 (Spectral weight equality). *For every state $|\psi\rangle$ in the SYK Hilbert space,*

$$\langle \psi | \mathcal{O}_m^\dagger \mathcal{O}_m | \psi \rangle = \langle \psi | \mathcal{O}_{A(m)}^\dagger \mathcal{O}_{A(m)} | \psi \rangle. \quad (14)$$

In particular, for the SYK ground state $|0\rangle$ and the spectral weight $w_m := \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle$,

$$w_m = w_{A(m)}. \quad (15)$$

Proof. From $\mathcal{O}_{A(m)} = -\mathcal{O}_m$,

$$\mathcal{O}_{A(m)}^\dagger \mathcal{O}_{A(m)} = (-\mathcal{O}_m)^\dagger (-\mathcal{O}_m) = \mathcal{O}_m^\dagger \mathcal{O}_m. \quad (16)$$

The expectation values in any state are therefore identical. □

Corollary 3.7 (No $1/N$ correction to the sum rule). *The sum rule*

$$w_m + w_{A(m)} = 2w_m \quad (17)$$

holds exactly, at every order in $1/N$. In particular, no coefficient c_1 appears, and the marginal kernel derivative k'_c does not contribute.

Proof. Immediate from Corollary 3.6. Since $w_m = w_{A(m)}$ is an operator-level identity, no expansion in $1/N$ is required and no correction term exists. $\square$

3.4 Vanishing bilinears for self-paired modes

Proposition 3.8 (Self paired modes vanish). *For $m \in \{(0,4), (4,0)\}$, the bilinear operator vanishes:*

$$\mathcal{O}_{(0,4)} = \mathcal{O}_{(4,0)} = 0. \quad (18)$$

Proof. For $m = (0,4)$: $v_{(0,4)}(x) = e^{i\pi \cdot 4 \cdot x_2/4} = e^{i\pi x_2} = (-1)^{x_2}$, which is real-valued for all $x \in \Lambda$. For $m = (4,0)$: $v_{(4,0)}(x) = (-1)^{x_1}$, also real-valued. For any real-valued wavefunction, $c_{ab}^m = 2\text{Im}(\text{real} \times \text{real}) = 0$. Therefore $\mathcal{O}_m = 0$. Alternatively, from the operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ with $A(m) = m$ for these two modes, we obtain $\mathcal{O}_m = -\mathcal{O}_m$, hence $\mathcal{O}_m = 0$. $\square$

4 Numerical Verification

4.1 Setup

We verify Corollary 3.6 for the SYK₄ model at $N = 16$ Majorana fermions, with the embedding $\iota(a) = (\lfloor a/8 \rfloor, a \bmod 8)$ (row embedding). We average over 10 disorder realizations (seeds 42–51).

For each seed, we diagonalize the $2^{N/2} = 256$ -dimensional Hamiltonian and compute the ground-state spectral weight $w_m = \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle$ for all 14 modes. The ensemble-averaged value $\overline{w_m} = \frac{1}{10} \sum_{\text{seeds}} w_m$ is reported.

4.2 Result

Table 1: Ensemble-averaged spectral weights by antipode pair.

Mode m	$\overline{w_m}$	Antipode $A(m)$	$\overline{w_{A(m)}}$
(2, 2)	56.65602	(6, 6)	56.65602
(6, 6)	56.65602	(2, 2)	56.65602
(7, 3)	56.45111	(1, 5)	56.45111
(1, 5)	56.45111	(7, 3)	56.45111
(3, 1)	56.25919	(5, 7)	56.25919
(5, 7)	56.25919	(3, 1)	56.25919
(6, 2)	56.12567	(2, 6)	56.12567
(2, 6)	56.12567	(6, 2)	56.12567
(5, 1)	55.77021	(3, 7)	55.77021
(3, 7)	55.77021	(5, 1)	55.77021
(1, 3)	55.75561	(7, 5)	55.75561
(7, 5)	55.75561	(1, 3)	55.75561
(0, 4)	0	(0, 4)	0
(4, 0)	0	(4, 0)	0

Every antipodal pair has identical ensemble-averaged weight to all displayed digits, in agreement with the exact identity of Corollary 3.6.

4.3 On the conformal dimension of the bilinears

A separate numerical computation fits the scaling exponent of the bilinear two-point function $\langle \mathcal{O}_m^\dagger(\tau) \mathcal{O}_m(0) \rangle$ in the intermediate-time window $\tau \in [0.5, 3.0]$ at $N = 16$. The fitted value is

$$h_m^{\text{fit}} = 0.31 \pm 0.05. \quad (19)$$

We emphasize that this is a *suggestive numerical indicator*, not a determination of the true infrared conformal dimension. At $N = 16$, with a Hilbert space of dimension $2^8 = 256$, the intermediate-time window is dominated by finite-size effects, ultraviolet tails, and discrete level spacing. The fit is consistent with both the free UV value $h_m = 2\Delta_\psi = 1/2$ and the SYK IR value $h_m = 2\Delta_\psi = 1/4$, and cannot distinguish between them at this system size. What the fit does establish is that $h_m \neq 2$ to high confidence, and therefore the marginal-dimension L'Hôpital argument for a coefficient c_1 used in the earlier paper does not apply.

A larger- N computation ($N = 20$ or $N = 24$) with sparse diagonalization methods would be needed to sharpen this measurement. Such a computation is beyond the scope of the present paper; the main theorem does not depend on any specific value of h_m .

5 Discussion

5.1 Scope of the “no $1/N$ correction” claim

The statement that the antipode sum rule has no $1/N$ correction must be scoped precisely. It applies to the specific spectral weights $w_m = \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle$ of the bilinear operators constructed from the 14 torus modes.

For these specific quantities, the identity is exact because $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ holds as an operator identity at every realization. No expansion in $1/N$ is required, and no correction term exists. This is not a general non-renormalization theorem. It does not imply that other correlators, or other composite operators built from $\mathcal{O}_m$, have vanishing $1/N$ corrections. For example, anomalous dimensions of higher composite operators, three-point functions, or non-antipodal correlators are not constrained by the identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$, and may well have non-trivial $1/N$ corrections.

The precise claim is: *for the spectral weights of the bilinear operators at the 14 torus modes, the antipode sum rule is exact at every order in $1/N$.* Nothing more is claimed.

5.2 Why the earlier paper’s claim failed

The earlier paper [5] reported a specific per-realization visible/hidden labeling of the 14 modes at $N = 16$, seed 42, row embedding. The present work shows this is not consistent with the exact operator identity of Theorem 3.5: if $\mathcal{O}_{A(m)} = -\mathcal{O}_m$, then $\mathcal{O}_m^\dagger \mathcal{O}_m = \mathcal{O}_{A(m)}^\dagger \mathcal{O}_{A(m)}$, and the two modes of an antipodal pair have identical spectral weights in *every* state, including every low-energy state. No labeling of the pair as “visible” and “hidden” can be stable across realizations because the two modes are indistinguishable by construction.

Numerical check: across 10 disorder realizations with the top-7-by-weight criterion, no mode appears in the visible set of every realization, and 12 of 14 modes appear in the visible set of at least one realization. The labeling is unstable, confirming that no geometric or algebraic invariant distinguishes the two members of an antipodal pair.

5.3 The physical content of the theorem

The theorem states that the torus geometry at $\lambda = 4$ has an exact $\mathbb{Z}_2$ symmetry acting as complex conjugation on the plane-wave basis, and that this symmetry is realized on the SYK bilinears as the operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$. The physical content is:

1. The two members of an antipodal pair are *not* independent operators. They are negatives of each other.
2. Consequently, any observable that is quadratic in $\mathcal{O}_m$ (such as the spectral weight) cannot distinguish between them.
3. The symmetry is a property of the torus geometry alone, not of the SYK dynamics, and therefore holds for any Hamiltonian defined on the substrate.

This is a clean, exact statement. It does not require ensemble averages, $1/N$ expansions, or numerical approximation.

5.4 Falsification

The theorem makes the following falsifiable predictions:

1. For every realization of the SYK₄ ensemble and every state, $w_m = w_{A(m)}$ exactly. Any violation falsifies the theorem.
2. The bilinear weights of the self-paired modes $(0, 4)$ and $(4, 0)$ vanish. Any nonzero weight for these modes falsifies the theorem.
3. No geometric invariant of the torus distinguishes the two members of an antipodal pair. Any stable per-realization labeling falsifies the theorem.

6 Conclusion

We have proven the SYK-Antipode Theorem: for the 8×8 torus at $\lambda = 4$, the bilinear operators constructed from antipodal modes satisfy the exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ for every mode m and every embedding of Majorana indices into torus sites. The identity is exact, depends only on the torus geometry, and does not require the SYK Hamiltonian or any ensemble average. It implies $w_m = w_{A(m)}$ for every state and every realization, so the antipode sum rule is exact at all orders in $1/N$ with no coefficient c_1 and no k'_c contribution.

The result corrects an earlier claim of a per-realization visible/hidden labeling: no such labeling can exist, because the operator identity makes the two members of every antipodal pair indistinguishable in every observable quadratic in $\mathcal{O}_m$.

References

- [1] S. Sachdev and J. Ye, *Gapless spin-fluid ground state in a random quantum Heisenberg magnet*, Phys. Rev. Lett. **70**, 3339 (1993).
- [2] A. Kitaev, *A simple model of quantum holography*, Talks at KITP (2015).
- [3] J. Maldacena and D. Stanford, *Remarks on the Sachdev-Ye-Kitaev model*, Phys. Rev. D **94**, 106002 (2016).
- [4] V. Rosenhaus, *Uncolored Random Tensors, Melon Diagrams, and the SYK Models*, arXiv:1611.08915 (2016).
- [5] S. Preschutti, *The SYK-Antipode Correspondence on the 8×8 Torus*, Entropia Research Institute Preprint (2026). [Corrected by the present paper.]
- [6] S. Preschutti, *The WIN Paradigm Explained*, WIN Paradigm White Paper (2026).